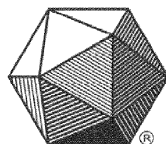


THE MATHEMATICAL ASSOCIATION OF AMERICA
American Mathematics Competitions



2013

AMC 8

(American Mathematics Contest 8)

INSTRUCTIONS

1. DO NOT OPEN THIS BOOKLET UNTIL YOUR PROCTOR TELLS YOU.
2. This is a twenty-five question multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
3. Mark your answer to each problem on the AMC 8 Answer Form with a #2 pencil. Check the blackened circles for accuracy and erase errors and stray marks completely. Only answers properly marked on the answer form will be graded.
4. There is no penalty for guessing. Your score on this test is the number of correct answers.
5. No aids are permitted other than scratch paper, graph paper, rulers, and erasers. No calculators are allowed. No problems on the test will *require* the use of a calculator.
6. Figures are not necessarily drawn to scale.
7. Before beginning the test, your proctor will ask you to record certain information on the answer form.
8. When your proctor gives the signal, begin working on the problems. You will have 40 minutes to complete the test.
9. When you finish the exam, *sign your name* in the space provided on the Answer Form.

The Committee on the American Mathematics Competitions reserves the right to re-examine students before deciding whether to grant official status to their scores. The Committee also reserves the right to disqualify all scores from a school if it determines that the required security procedures were not followed.

The American Mathematics Competitions (AMC)

The AMC 8 is for students grades 8 and under. It is a 25 question, 40 minute multiple choice test. Problems generally increase in difficulty as the exam progresses. Through 2007, calculators were permitted, but they will not be in the future. Unlike the AMC 10 and 12, no points are deducted for wrong answers.

2013 AMC 8 Problems

Problem 1

Amma wants to arrange her model cars in rows with exactly 6 cars in each row. She now has 23 model cars. What is the smallest number of additional cars she must buy in order to be able to arrange all her cars this way?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 2

A sign at the fish market says, "50% off, today only: half-pound packages for just \$3 per package." What is the regular price for a full pound of fish, in dollars?

- (A) 6 (B) 9 (C) 10 (D) 12 (E) 15

Problem 3

What is the value of $4 \cdot (-1 + 2 - 3 + 4 - 5 + 6 - 7 + \dots + 1000)$?

- (A) -10 (B) 0 (C) 1 (D) 500 (E) 2000

Problem 4

Eight friends ate at a restaurant and agreed to share the bill equally. Because Judi forgot her money, each of her seven friends paid an extra \$2.50 to cover her portion of the total bill. What was the total bill?

- (A) \$120 (B) \$128 (C) \$140 (D) \$144 (E) \$160

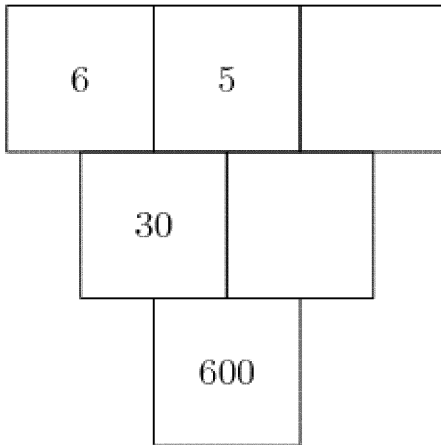
Problem 5

Hammie is in the 6th grade and weighs 106 pounds. His quadruplet sisters are tiny babies and weigh 5, 5, 6, and 8 pounds. Which is greater, the average (mean) weight of these five children or the median weight, and by how many pounds?

- (A) median, by 60 (B) median, by 20 (C) average, by 5 (D) average, by 15 (E) average, by 20

Problem 6

The number in each box below is the product of the numbers in the two boxes that touch it in the row above. For example, $30 = 6 \times 5$. What is the missing number in the top row?



- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Problem 7

Trey and his mom stopped at a railroad crossing to let a train pass. As the train began to pass, Trey counted 6 cars in the first 10 seconds. It took the train 2 minutes and 45 seconds to clear the crossing at a constant speed. Which of the following was the most likely number of cars in the train?

- (A) 60 (B) 80 (C) 100 (D) 120 (E) 140

Problem 8

A fair coin is tossed 3 times. What is the probability of at least two consecutive heads?

- (A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{3}{8}$ (D) $\frac{1}{2}$ (E) $\frac{3}{4}$

Problem 9

The Incredible Hulk can double the distance he jumps with each succeeding jump. If his first jump is 1 meter, the second jump is 2 meters, the third jump is 4 meters, and so on, then on which jump will he first be able to jump more than 1 kilometer?

- (A) 9th (B) 10th (C) 11th (D) 12th (E) 13th

Problem 10

What is the ratio of the least common multiple of 180 and 594 to the greatest common factor of 180 and 594?

- (A) 110 (B) 165 (C) 330 (D) 625 (E) 660

Problem 11

Ted's grandfather used his treadmill on 3 days this week. He went 2 miles each day. On Monday he jogged at a speed of 5 miles per hour. He walked at the rate of 3 miles per hour on Wednesday and at 4 miles per hour on Friday. If Grandfather had always walked at 4 miles per hour, he would have spent less time on the treadmill. How many minutes less?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 12

At the 2013 Winnebago County Fair a vendor is offering a "fair special" on sandals. If you buy one pair of sandals at the regular price of \$50, you get a second pair at a 40% discount, and a third pair at half the regular price. Javier took advantage of the "fair special" to buy three pairs of sandals. What percentage of the \$150 regular price did he save?

- (A) 25 (B) 30 (C) 33 (D) 40 (E) 45

Problem 13

When Clara totaled her scores, she inadvertently reversed the units digit and the tens digit of one score. By which of the following might her incorrect sum have differed from the correct one?

- (A) 45 (B) 46 (C) 47 (D) 48 (E) 49

Problem 14

Abe holds 1 green and 1 red jelly bean in his hand. Bea holds 1 green, 1 yellow, and 2 red jelly beans in her hand. Each randomly picks a jelly bean to show the other. What is the probability that the colors match?

- (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{3}{8}$ (D) $\frac{1}{2}$ (E) $\frac{2}{3}$

Problem 15

If $3^p + 3^4 = 90$, $2^r + 44 = 76$, and $5^3 + 6^s = 1421$, what is the product of p , r , and s ?

- (A) 27 (B) 40 (C) 50 (D) 70 (E) 90

Problem 16

A number of students from Fibonacci Middle School are taking part in a community service project. The ratio of 8th-graders to 6th-graders is 5 : 3, and the ratio of 8th-graders to 7th-graders is 8 : 5. What is the smallest number of students that could be participating in the project?

- (A) 16 (B) 40 (C) 55 (D) 79 (E) 89

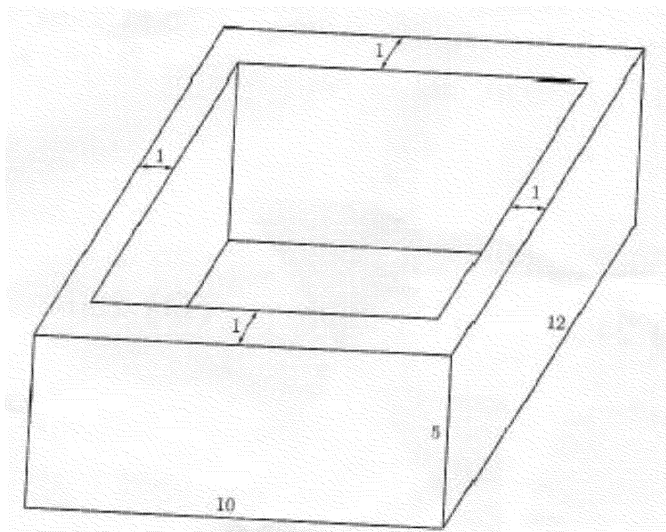
Problem 17

The sum of six consecutive positive integers is 2013. What is the largest of these six integers?

- (A) 335 (B) 338 (C) 340 (D) 345 (E) 350

Problem 18

Isabella uses one-foot cubical blocks to build a rectangular fort that is 12 feet long, 10 feet wide, and 5 feet high. The floor and the four walls are all one foot thick. How many blocks does the fort contain?



- (A) 204 (B) 280 (C) 320 (D) 340 (E) 600

Problem 19

Bridget, Cassie, and Hannah are discussing the results of their last math test. Hannah shows Bridget and Cassie her test, but Bridget and Cassie don't show theirs to anyone. Cassie says, 'I didn't get the lowest score in our class,' and Bridget adds, 'I didn't get the highest score.' What is the ranking of the three girls from highest to lowest?

- (A) Hannah, Cassie, Bridget (B) Hannah, Bridget, Cassie
(C) Cassie, Bridget, Hannah (D) Cassie, Hannah, Bridget
(E) Bridget, Cassie, Hannah

Problem 20

A 1×2 rectangle is inscribed in a semicircle with longer side on the diameter. What is the area of the semicircle?

- (A) $\frac{\pi}{2}$ (B) $\frac{2\pi}{3}$ (C) π (D) $\frac{4\pi}{3}$ (E) $\frac{5\pi}{3}$

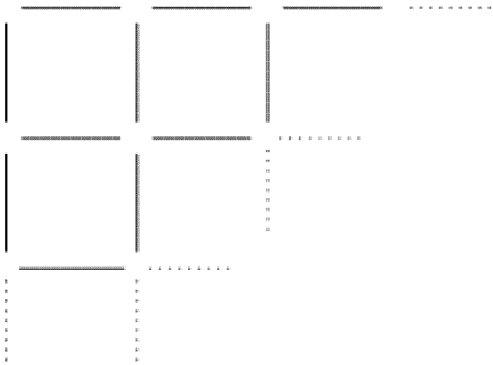
Problem 21

Samantha lives 2 blocks west and 1 block south of the southwest corner of City Park. Her school is 2 blocks east and 2 blocks north of the northeast corner of City Park. On school days she bikes on streets to the southwest corner of City Park, then takes a diagonal path through the park to the northeast corner, and then bikes on streets to school. If her route is as short as possible, how many different routes can she take?

- (A) 3 (B) 6 (C) 9 (D) 12 (E) 18

Problem 22

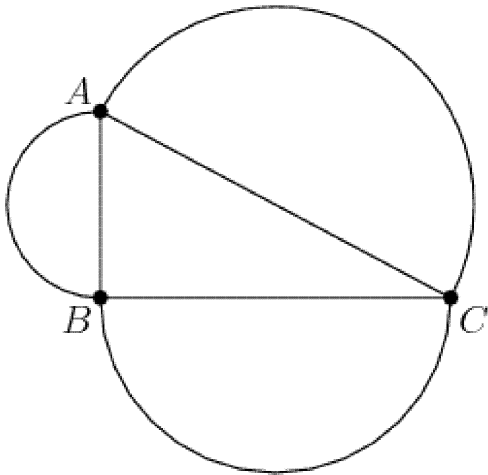
Toothpicks are used to make a grid that is 60 toothpicks long and 32 toothpicks wide. How many toothpicks are used altogether?



- (A) 1920 (B) 1952 (C) 1980 (D) 2013 (E) 3932

Problem 23

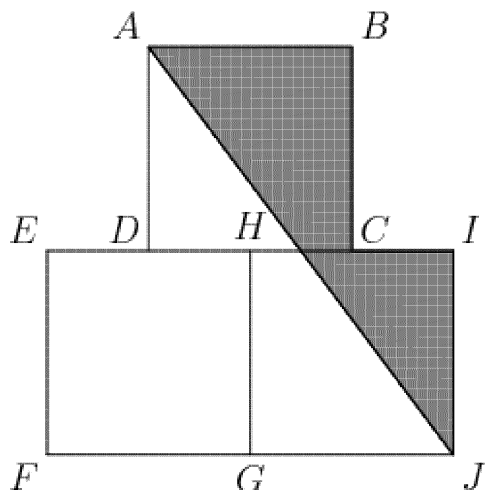
Angle ABC of $\triangle ABC$ is a right angle. The sides of $\triangle ABC$ are the diameters of semicircles as shown. The area of the semicircle on $\overline{AB}$ equals 8π , and the arc of the semicircle on $\overline{AC}$ has length 8.5π . What is the radius of the semicircle on $\overline{BC}$?



- (A) 7 (B) 7.5 (C) 8 (D) 8.5 (E) 9

Problem 24

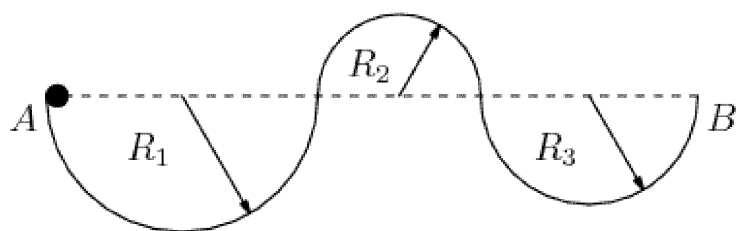
Squares $ABCD$, $EFGH$, and $GHIJ$ are equal in area. Points C and D are the midpoints of sides IH and HE , respectively. What is the ratio of the area of the shaded pentagon $AJICB$ to the sum of the areas of the three squares?



- (A) $\frac{1}{4}$ (B) $\frac{7}{24}$ (C) $\frac{1}{3}$ (D) $\frac{3}{8}$ (E) $\frac{5}{12}$

Problem 25

A ball with diameter 4 inches starts at point A to roll along the track shown. The track is comprised of 3 semicircular arcs whose radii are $R_1 = 100$ inches, $R_2 = 60$ inches, and $R_3 = 80$ inches, respectively. The ball always remains in contact with the track and does not slip. What is the distance the center of the ball travels over the course from A to B?



- (A) 238π (B) 240π (C) 260π (D) 280π (E) 500π

2013 AMC 8 Problems/Problem 13

Problem

When Clara totaled her scores, she inadvertently reversed the units digit and the tens digit of one score. By which of the following might her incorrect sum have differed from the correct one?

- (A) 45 (B) 46 (C) 47 (D) 48 (E) 49

Solution

Let the two digits be a and b .

The correct score was $10a + b$. Clara misinterpreted it as $10b + a$. The difference between the two is $|9a - 9b|$ which factors into $|9(a - b)|$. Therefore, since the difference is a multiple of 9, the only answer choice that is a multiple of 9 is **(A) 45**.

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2013 AMC 8 Answer Key

1. **(A)** 1
2. **(D)** 12
3. **(E)** 2000
4. **(C)** 140
5. **(E)** average, by 20
6. **(C)** 4
7. **(C)** 100
8. **(C)** $\frac{3}{8}$
9. **(C)** 11th
10. **(C)** 330
11. **(D)** 4
12. **(B)** 30
13. **(A)** 45
14. **(C)** $\frac{3}{8}$
15. **(B)** 40
16. **(E)** 89
17. **(B)** 338
18. **(B)** 280
19. **(D)** Cassie, Hannah, Bridget
20. **(C)** π
21. **(E)** 18
22. **(E)** 3932
23. **(B)** 7.5
24. **(C)** $\frac{1}{3}$
25. **(A)** 238π

2013 AMC 8 Problems/Problem 13

Problem

When Clara totaled her scores, she inadvertently reversed the units digit and the tens digit of one score. By which of the following might her incorrect sum have differed from the correct one?

- (A) 45 (B) 46 (C) 47 (D) 48 (E) 49

Solution

Let the two digits be a and b .

The correct score was $10a + b$. Clara misinterpreted it as $10b + a$. The difference between the two is $|9a - 9b|$ which factors into $|9(a - b)|$. Therefore, since the difference is a multiple of 9, the only answer choice that is a multiple of 9 is (A) 45.

2013 AMC 8 Problems/Problem 16

Problem

A number of students from Fibonacci Middle School are taking part in a community service project. The ratio of 8th-graders to 6th-graders is 5 : 3, and the ratio of 8th-graders to 7th-graders is 8 : 5. What is the smallest number of students that could be participating in the project?

- (A) 16 (B) 40 (C) 55 (D) 79 (E) 89

Solution

Solution 1: Algebra

We multiply the first ratio by 8 on both sides, and the second ratio by 5 to get the same number for 8th graders, in order that we can put the two ratios together:

$$5 : 3 = 5(8) : 3(8) = 40 : 24$$

$$8 : 5 = 8(5) : 5(5) = 40 : 25$$

Therefore, the ratio of 8th graders to 7th graders to 6th graders is 40 : 25 : 24. Since the ratio is in lowest terms, the smallest number of students participating in the project is $40 + 25 + 24 =$ (E) 89.

Solution 2: Fakesolving

The number of 8th graders has to be a multiple of 8 and 5, so assume it is 40 (the smallest possibility).

Then there are $40 * \frac{3}{5} = 24$ 6th graders and $40 * \frac{5}{8} = 25$ 7th graders. The numbers of students is $40 + 24 + 25 =$ (E) 89

2013 AMC 8 Problems/Problem 17

Problem

The sum of six consecutive positive integers is 2013. What is the largest of these six integers?

- (A) 335 (B) 338 (C) 340 (D) 345 (E) 350

Solution 1

The mean of these numbers is $\frac{\frac{2013}{3}}{2} = \frac{671}{2} = 335.5$. Therefore the numbers are 333, 334, 335, 336, 337, 338, so the answer is **(B) 338**.

Solution 2

Let the 4th number be x . Then our desired number is $x + 2$.

Our integers are $x - 3, x - 2, x - 1, x, x + 1, x + 2$, so we have that

$$6x - 3 = 2013 \implies x = \frac{2016}{6} = 336 \implies x + 2 = \mathbf{(B) 338}.$$

Solution 3

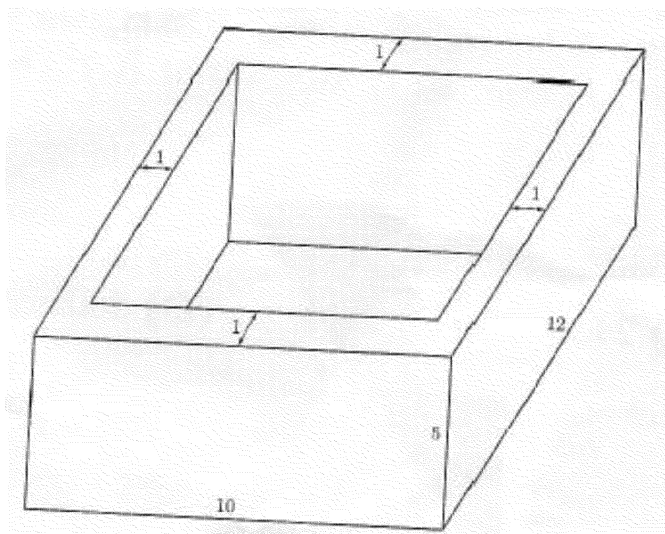
Let the first term be x . Our integers are $x, x + 1, x + 2, x + 3, x + 4, x + 5$. We have,

$$6x + 15 = 2013 \implies x = 333 \implies x + 5 = \mathbf{(B) 338}$$

2013 AMC 8 Problems/Problem 18

Problem

Isabella uses one-foot cubical blocks to build a rectangular fort that is 12 feet long, 10 feet wide, and 5 feet high. The floor and the four walls are all one foot thick. How many blocks does the fort contain?



--Arpanliku 16:22, 27 November 2013 (EST) Courtesy of Lord.of.AMC

- (A) 204 (B) 280 (C) 320 (D) 340 (E) 600

Solution 1

There are $10 \cdot 12 = 120$ cubes on the base of the box. Then, for each of the 4 layers above the bottom (as since each cube is 1 foot by 1 foot by 1 foot and the box is 5 feet tall, there are 4 feet left), there are $9 + 11 + 9 + 11 = 40$ cubes. Hence, the answer is $120 + 4 \cdot 40 = \boxed{\text{(B)} 280}$.

Solution 2

We can just calculate the volume of the prism that was cut out of the original $12 \times 10 \times 5$ box. Each interior side of the fort will be 2 feet shorter than each side of the outside. Since the floor is 1 foot, the height will be 4 feet. So the volume of the interior box is $10 \times 8 \times 4 = 320 \text{ ft}^3$.

The volume of the original box is $12 \times 10 \times 5 = 600 \text{ ft}^3$. Therefore, the number of blocks contained in the fort is $600 - 320 = \boxed{\text{(B)} 280}$.

2013 AMC 8 Problems/Problem 19

Problem

Bridget, Cassie, and Hannah are discussing the results of their last math test. Hannah shows Bridget and Cassie her test, but Bridget and Cassie don't show theirs to anyone. Cassie says, 'I didn't get the lowest score in our class,' and Bridget adds, 'I didn't get the highest score.' What is the ranking of the three girls from highest to lowest?

- (A) Hannah, Cassie, Bridget (B) Hannah, Bridget, Cassie
(C) Cassie, Bridget, Hannah (D) Cassie, Hannah, Bridget
(E) Bridget, Cassie, Hannah

Solution

If Hannah did better than Cassie, there would be no way she could know for sure that she didn't get the lowest score in the class. Therefore, Hannah did worse than Cassie. Similarly, if Hannah did worse than Bridget, there is no way Bridget could have known that she didn't get the highest in the class. Therefore, Hannah did better than Bridget, so our order is $\boxed{\text{(D)} \text{ Cassie, Hannah, Bridget }}$.

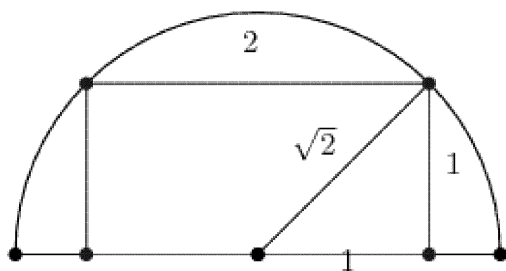
2013 AMC 8 Problems/Problem 20

Problem

A 1×2 rectangle is inscribed in a semicircle with longer side on the diameter. What is the area of the semicircle?

- (A) $\frac{\pi}{2}$ (B) $\frac{2\pi}{3}$ (C) π (D) $\frac{4\pi}{3}$ (E) $\frac{5\pi}{3}$

Solution



A semicircle has symmetry, so the center is exactly at the midpoint of the 2 side on the rectangle, making the radius, by the Pythagorean Theorem, $\sqrt{1^2 + 1^2} = \sqrt{2}$. The area is $\frac{2\pi}{2} = \boxed{\text{(C)} \pi}$.

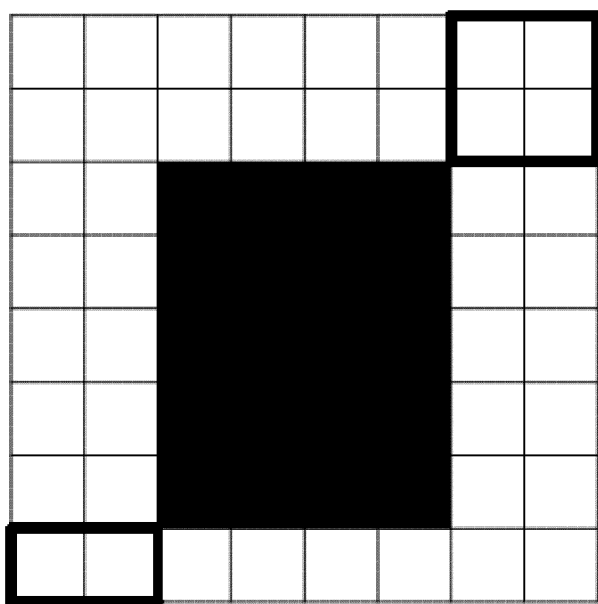
2013 AMC 8 Problems/Problem 21

Problem

Samantha lives 2 blocks west and 1 block south of the southwest corner of City Park. Her school is 2 blocks east and 2 blocks north of the northeast corner of City Park. On school days she bikes on streets to the southwest corner of City Park, then takes a diagonal path through the park to the northeast corner, and then bikes on streets to school. If her route is as short as possible, how many different routes can she take?

- (A) 3 (B) 6 (C) 9 (D) 12 (E) 18

Solution

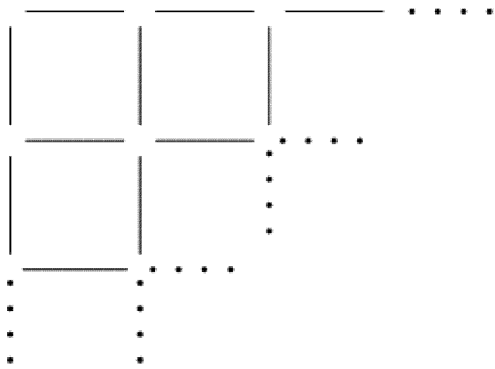


The number of ways to get from Samantha's house to City Park is $\binom{3}{1} = 3$, and the number of ways to get from City Park to school is $\binom{4}{2} = 6$. Since there's one way to go through City Park (just walking straight through), the number of different ways to go from Samantha's house to City Park to school is $3 \cdot 6 = \boxed{\text{(E)} 18}$.

2013 AMC 8 Problems/Problem 22

Problem

Toothpicks are used to make a grid that is 60 toothpicks long and 32 toothpicks wide. How many toothpicks are used altogether?



- (A) 1920 (B) 1952 (C) 1980 (D) 2013 (E) 3932

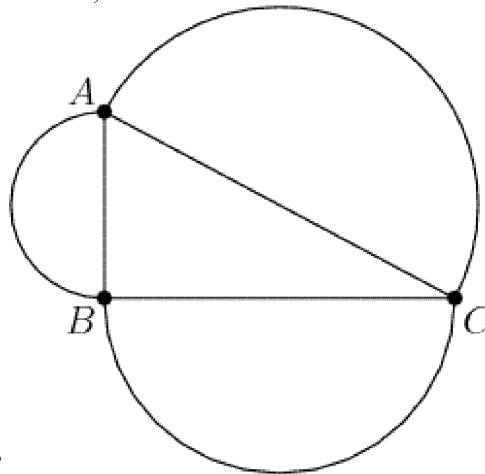
Solution

There are 61 vertical columns with a length of 32 toothpicks, and there are 33 horizontal rows with a length of 60 toothpicks. An effective way to verify this is to try a small case, i.e. a 2×3 grid of toothpicks. Thus, our answer is $61 \cdot 32 + 33 \cdot 60 = \boxed{\text{(E)} 3932}$.

2013 AMC 8 Problems/Problem 23

Problem

Angle ABC of $\triangle ABC$ is a right angle. The sides of $\triangle ABC$ are the diameters of semicircles as shown. The area of the semicircle on $\overline{AB}$ equals 8π , and the arc of the semicircle on $\overline{AC}$ has length 8.5π . What is



the radius of the semicircle on $\overline{BC}$?

- (A) 7 (B) 7.5 (C) 8 (D) 8.5 (E) 9

Solution 1

If the semicircle on AB were a full circle, the area would be 16π . Therefore the diameter of the first circle is 8. The arc of the largest semicircle would normally have a complete diameter of 17. The

Pythagorean theorem says that the other side has length 15, so the radius is $\boxed{\text{(B)} 7.5}$.

Solution 2

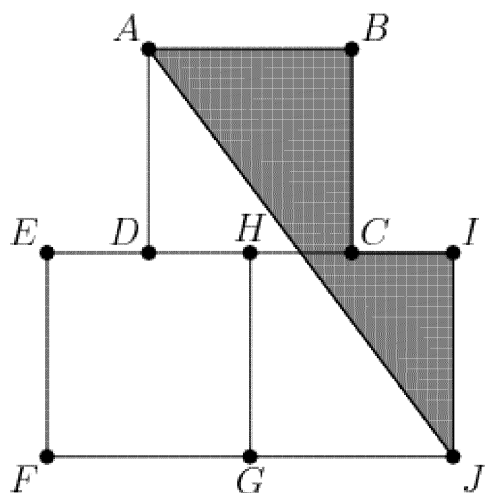
We go as in Solution 1, finding the diameter of the circle on AC and AB . Then, an extended version of the theorem says that the sum of the semicircles on the left is equal to the biggest one, so the area of the largest is $\frac{289\pi}{8}$, and the middle one is $\frac{289\pi}{8} - \frac{64\pi}{8} = \frac{225\pi}{8}$, so the radius is $\frac{15}{2} = \boxed{\text{(B)} 7.5}$.

2013 AMC 8 Problems/Problem 24

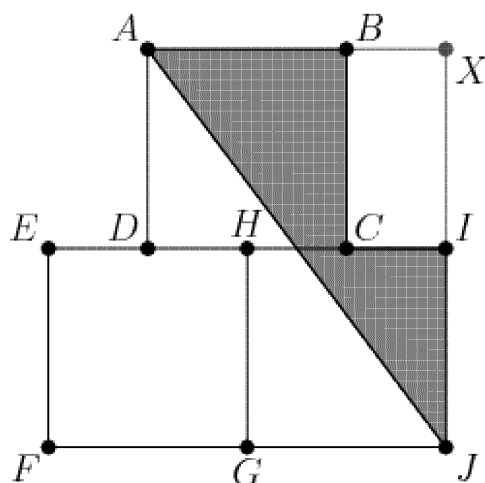
Problem

Squares $ABCD$, $EFGH$, and $GHIJ$ are equal in area. Points C and D are the midpoints of sides IH and HE , respectively. What is the ratio of the area of the shaded pentagon $AJICB$ to the sum of the areas of the three squares?

- (A) $\frac{1}{4}$ (B) $\frac{7}{24}$ (C) $\frac{1}{3}$ (D) $\frac{3}{8}$ (E) $\frac{5}{12}$



Solution 1

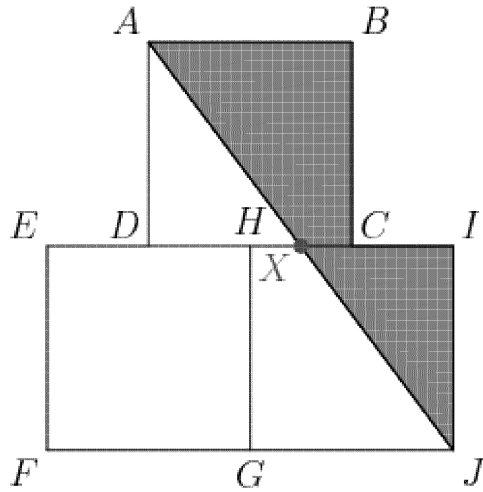


First let $s = 2$ (where s is the side length of the squares) for simplicity. We can extend IJ until it hits the extension of AB . Call this point X . The area of triangle AXJ then is $\frac{3 \cdot 4}{2}$. The area of rectangle $BXIC$

is $2 \cdot 1 = 2$. Thus, our desired area is $6 - 2 = 4$. Now, the ratio of the shaded area to the combined area

of the three squares is $\frac{4}{3 \cdot 2^2} = \boxed{\text{(C)} \frac{1}{3}}$.

Solution 2



Let the side length of each square be 1.

Let the intersection of AJ and EI be X .

Since $[ABCD] = [GHIJ]$, $AD = IJ$. Since $\angle IXJ$ and $\angle AXD$ are vertical angles, they are congruent. We also have $\angle JIH \cong \angle ADC$ by definition.

So we have $\triangle ADX \cong \triangle JIX$ by AAS congruence. Therefore, $DX = JX$.

Since C and D are midpoints of sides, $DH = CJ = \frac{1}{2}$. This combined with $DX = JX$ yields $HX = CX = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

The area of trapezoid $ABCX$ is $\frac{1}{2}(AB + CX)(BC) = \frac{1}{2} \times \frac{5}{4} \times 1 = \frac{5}{8}$.

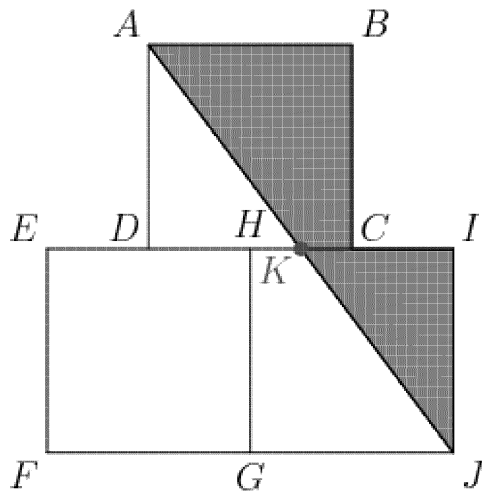
The area of triangle JIX is $\frac{1}{2} \times XJ \times IJ = \frac{1}{2} \times \frac{3}{4} \times 1 = \frac{3}{8}$.

So the area of the pentagon $AJICB$ is $\frac{3}{8} + \frac{5}{8} = 1$.

The area of the 3 squares is $1 \times 3 = 3$.

Therefore, $\frac{[AJICB]}{[ABCIJFED]} = \boxed{\text{(C)} \frac{1}{3}}$.

Solution 3



Let the intersection of AJ and EI be K .

Now we have $\triangle ADK$ and $\triangle KIJ$.

Because both triangles have a side on congruent squares therefore $AD \cong IJ$.

Because $\angle AKD$ and $\angle JKI$ are vertical angles $\angle AKD \cong \angle JKI$.

Also both $\angle ADK$ and $\angle JIK$ are right angles so $\angle ADK \cong \angle JIK$.

Therefore by AAS (Angle, Angle, Side) $\triangle ADK \cong \triangle KIJ$.

Then translating/rotating the shaded $\triangle JIK$ into the position of $\triangle ADK$

So the shaded area now completely covers the square $ABCD$

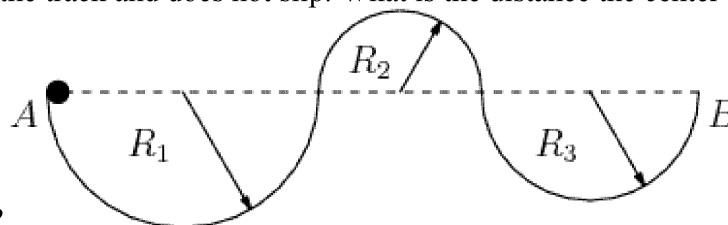
Set the area of a square as x

Therefore, $\frac{x}{3x} = \boxed{(C) \frac{1}{3}}$.

2013 AMC 8 Problems/Problem 25

Problem

A ball with diameter 4 inches starts at point A to roll along the track shown. The track is comprised of 3 semicircular arcs whose radii are $R_1 = 100$ inches, $R_2 = 60$ inches, and $R_3 = 80$ inches, respectively. The ball always remains in contact with the track and does not slip. What is the distance the center of the

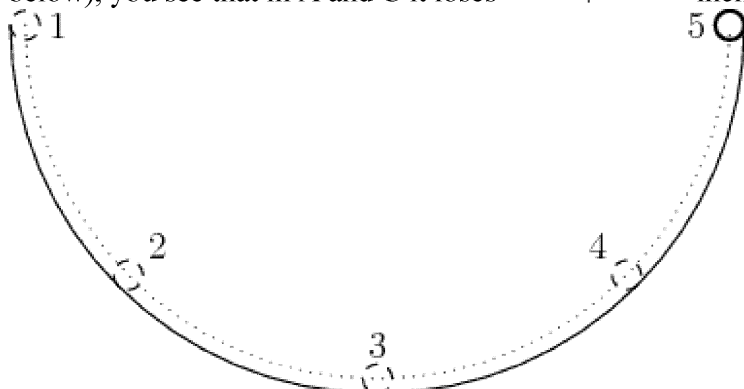


ball travels over the course from A to B?

- (A) 238π (B) 240π (C) 260π (D) 280π (E) 500π

Solution 1

The radius of the ball is 2 inches. If you think about the ball rolling or draw a path for the ball (see figure below), you see that in A and C it loses $2\pi * 2/2 = 2\pi$ inches, and it gains 2π inches on B.



So, the departure from the length of the track means that the answer is $\frac{200 + 120 + 160}{2} * \pi + (-2 - 2 + 2) * \pi = 240\pi - 2\pi = \boxed{\text{(A)} 238\pi}$.

Solution 2

The total length of all of the arcs is $100\pi + 80\pi + 60\pi = 240\pi$. Since we want the path from the center, the actual distance will be shorter. Therefore, the only answer choice less than 240π is $\boxed{\text{(A)} 238\pi}$. This solution may be invalid because the actual distance can be longer if the path the center travels is on the outside of the curve, as it is in the middle bump.